UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 2 2023/2024

SECV2113 HUMAN COMPUTER INTERACTION

SECTION 01

PROJECT 2a

INDUSTRY INNOVATION AND INFRASTRUCTURE

LECTURER: DR. IZYAN IZZATI KAMSANI

GROUP 9

Student Name	Matric No.
LAM YOKE YU	A23CS0233
GOE JIE YING	A23CS0224
TEH RU QIAN	A23CS0191
TAN YI YA	A23CS0187

Task Analysis

Introduction

During the shopping process at the mall, stakeholders including busy professionals, inventory managers, and seniors or disabled people utilize self-checkout technology for seamless transactions. They begin with selecting items, browsing through the store's aisles to find their desired products efficiently. Next, they effortlessly scan barcodes using either barcode scanner or manual entry options provided by the self-checkout system. Finally, making payments is a breeze, with a variety of options available, including contactless payments for speed and convenience. This streamlined process ensures a smooth shopping experience for all stakeholders, catering to their diverse needs and preferences.

HTA Derivation

Task 1 - Select Items

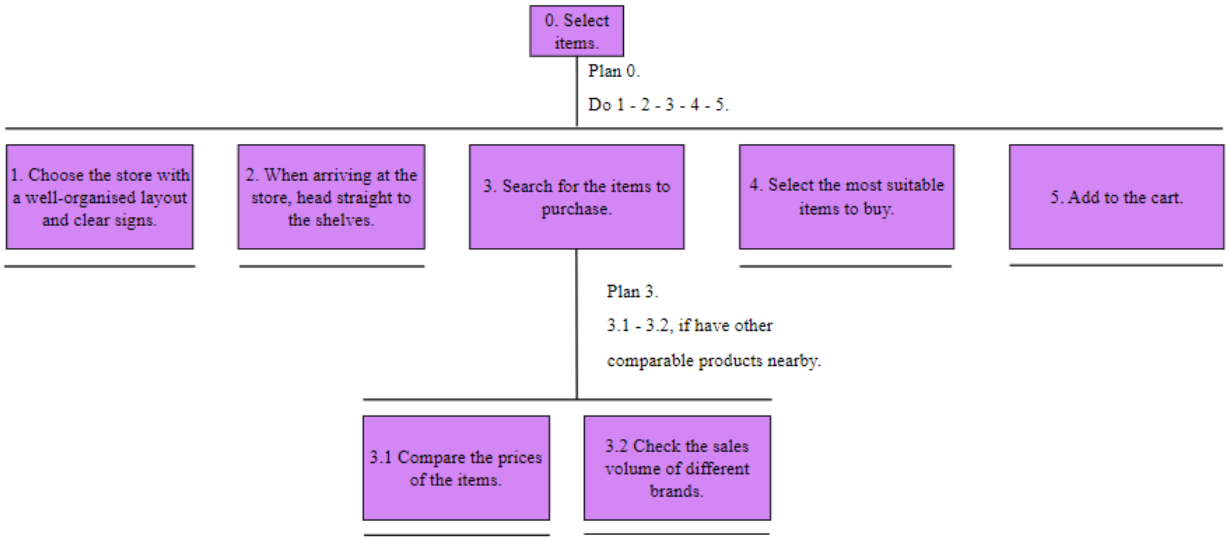
a. Busy Professionals

Video link : https://youtu.be/nvLW_7PVioo?si=0EzlkoNdB4GVJYbf

0. Select items.
1. Choose the store with a well-organised layout and clear signs.
2. When arriving at the store, head straight to the shelves.
3. Search for the items to purchase.
    - 3.1 Compare the prices of the items.
    - 3.2 Check the sales volume of different brands.
4. Select the most suitable items to buy.
5. Add to the cart.

Plan 0: Do 1, 2, 3, 4 then 5.

Plan 3: Do 3.1 and 3.2, if there are other comparable products nearby



b. Inventory Managers

Video link : https://youtu.be/9YpXHxNdloI?si=xLa5FtNsMaVrxRFz

0. Choose items.

1. Generate shopping lists based on the inventory needs.

2. Search and choose the location.

3. Double-check product details against inventory records to prevent discrepancies.

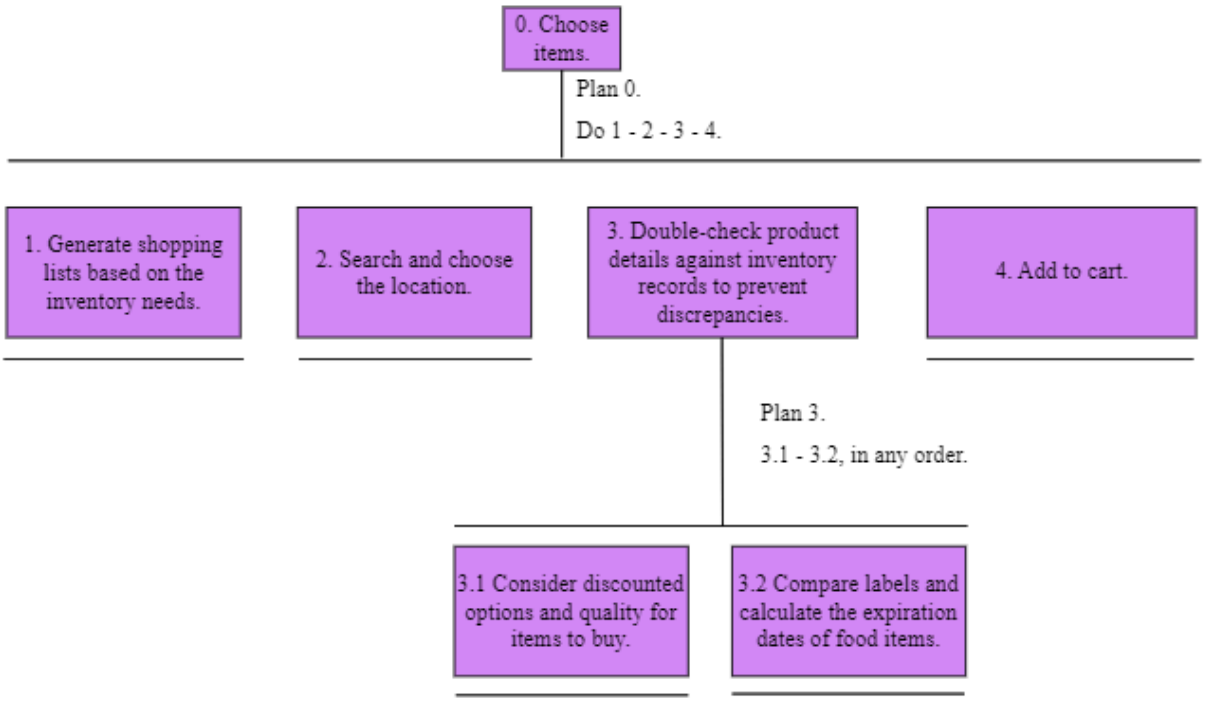
    3.1 Consider discounted options and quality for items to buy.

    3.2 Compare labels and calculate the expiration dates of food items.

4. Add to cart.

Plan 0: Do 1,2, 3 then 4.

Plan 3: Do 3.1 or 3.2 in any order.



c. Disabled People

Video link : https://youtu.be/-boH4jJJifE?si=8yTZRJLBw006j0NW

0. Buy items.

1. Write a shopping list.

2. Choose stores that are nearby his house.

3. Find the items.

    3.1 Put the selected items into the trolly.

    3.2 Seek assistance from store staff for item location.

4. Review each item.

    4.1 Check product labels, barcodes, or packaging for any discrepancies.

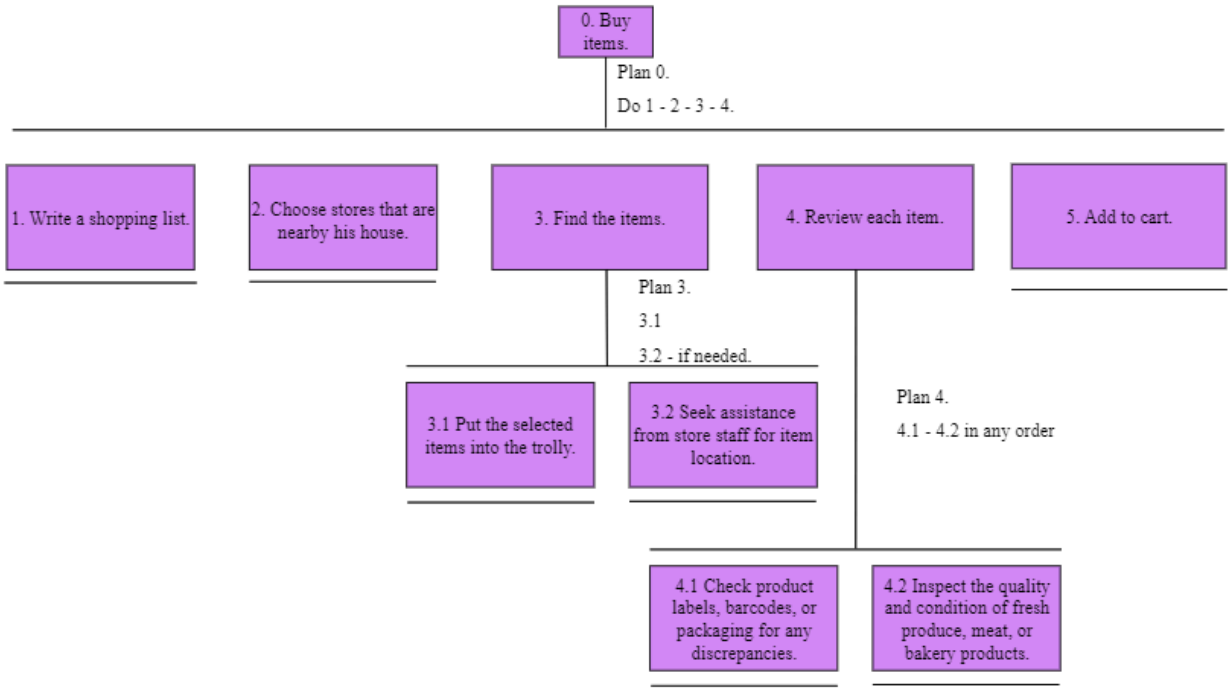
    4.2 Inspect the quality and condition of fresh produce, meat, or bakery products.

5. Add to cart.

Plan 0: Do 1, 2, 3, then 4.

Plan 3: Do 3.2, if needed.

Plan 4: Do 4.1 or 4.2 in any order.



Findings from the HTAs for Task 1:

- Busy professionals, inventory managers and disabled people have encountered no issues when selecting items.
- The store shelves are neatly arranged according to the types of items, allowing users to quickly find what they need.
- Even if unable to locate an item or if an item is placed too high for customers to reach, they can easily find assistance from nearby staff or other customers.

Task 2 - Scan Barcodes

a. Busy Professionals

Video link : https://youtu.be/k-cZCOqiM94?si=KM6lSx-wmOwAeJXj

0. Scan Barcodes.

1. Tap the screen to start.

2. Confirm membership status.

3. Scan to purchase items.

    3.1 Look for the barcode on the items.

    3.2 Put the barcode near the scanner to scan.

    3.3 Place the scanned items in the bagging area.

4. Check the lists of items purchased.

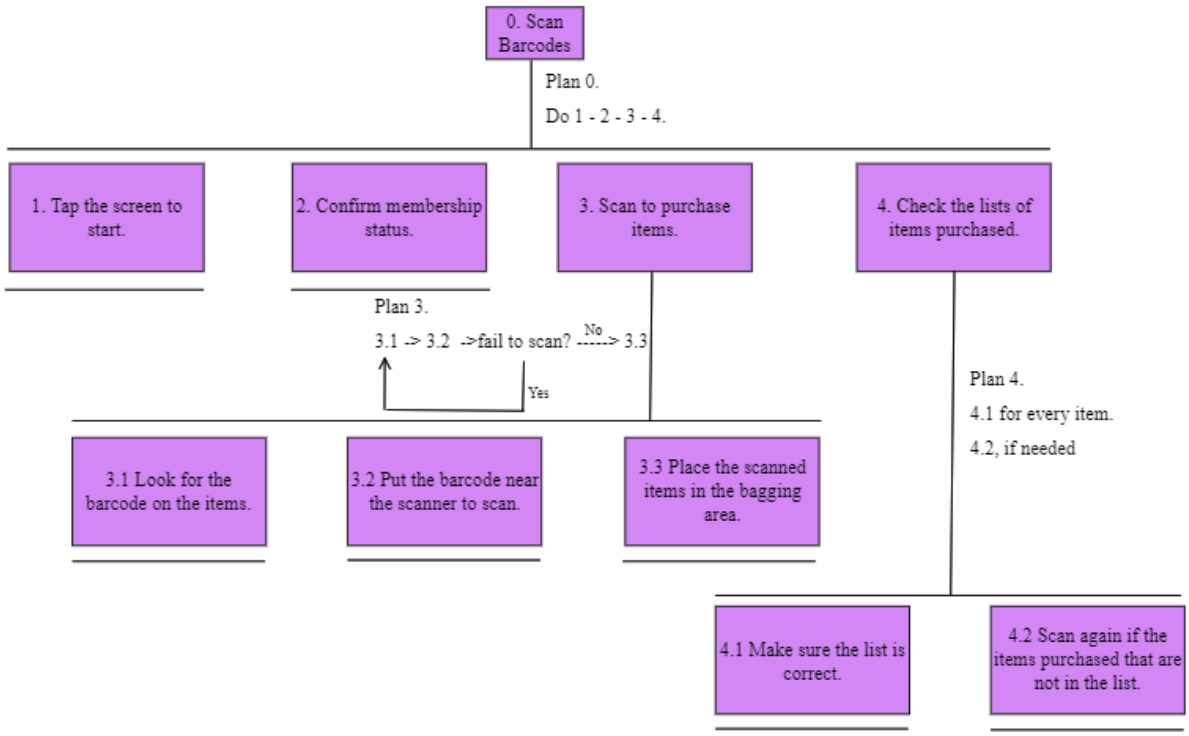
    4.1 Make sure the list is correct.

    4.2 Scan again if the items purchased that are not in the list.

Plan 0: Do 1, 2, 3 then 4.

Plan 3: Do 3.1 and (3.2, if fail to scan, back to 3.1 again), then 3.3.

Plan 4: Do 4.1 for every item, then 4.2 if needed.



b. Inventory Managers

Video link : https://youtu.be/0J0ujyDKXN0?si=ROVL_9PIarke95vh

0. Scan items.

1. Tap the screen to begin the checkout process.

2. Confirm membership status.

3. Start scanning items.

    3.1 Place the items near the barcode scanner.

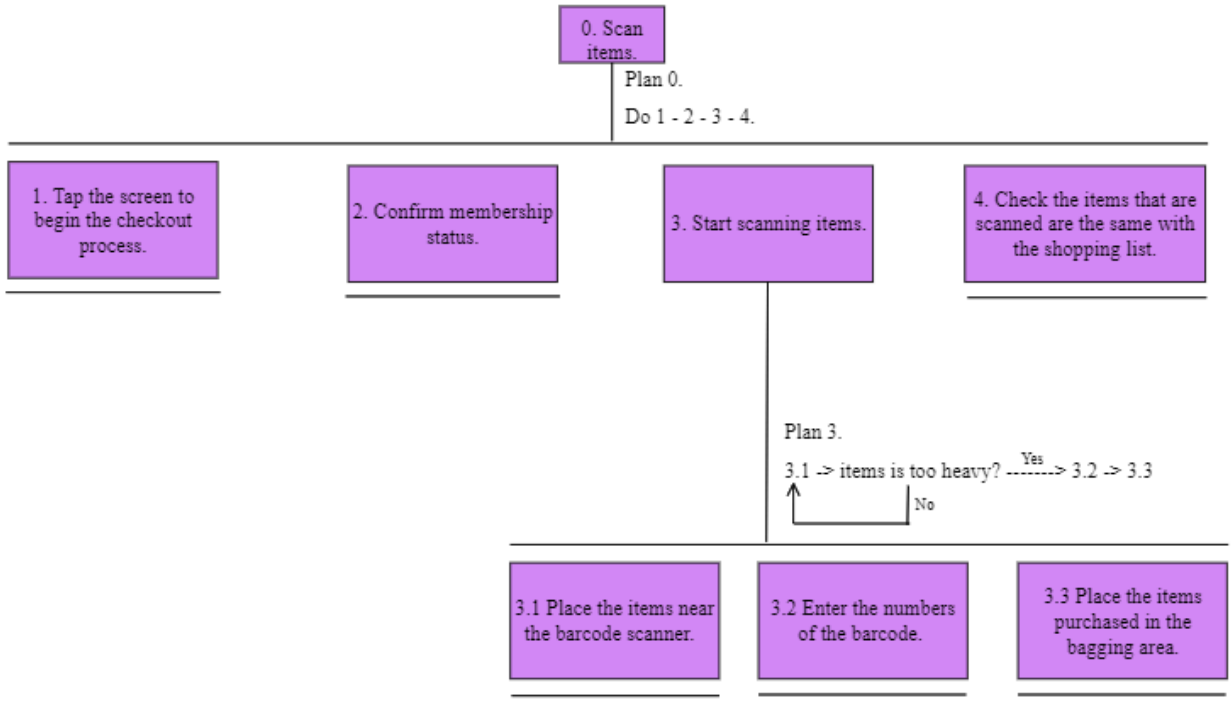
    3.2 Enter the numbers of the barcode.

    3.3 Place the items purchased in the bagging area.

4. Check the items that are scanned are the same with the shopping list.

Plan 0: Do 1, 2, 3 then 4.

Plan 3: Do (3.1 repeatedly), 3.2, if the item is too heavy and hard to place near the scanner then 3.3.



c. Disabled People

Video link : https://youtu.be/XZ3a4N77Wmg?si=Wqbx26xqDOr3WhoI

0. Buy items.

1. Place the items on the counter.

2. Start by tapping the screen.

3. Confirm membership status.

4. Begin to scan the barcode.

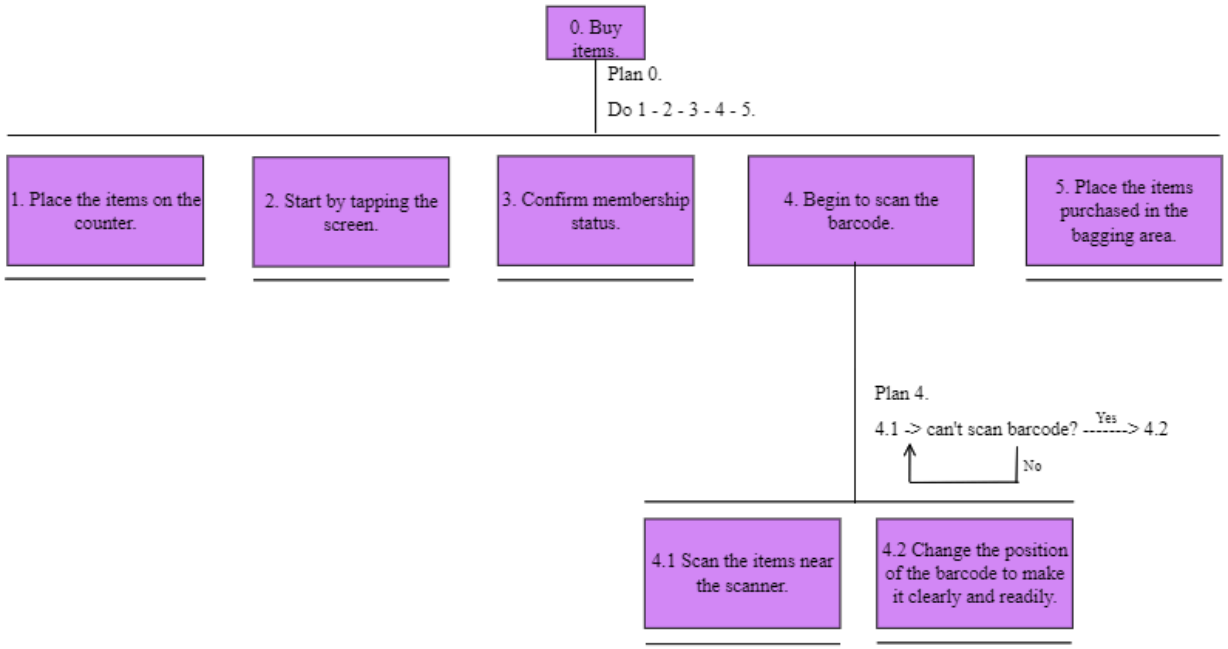
    4.1 Scan the items near the scanner.

    4.2 Change the position of the barcode to make it clearly and readily.

5. Place the items purchased in the bagging area.

Plan 0: Do 1, 2, 3, 4 then 5.

Plan 4: Do 4.1 and (4.2 repeatedly, if the scanner can’t scan the barcode).



Findings from the HTAs for Task 2:

- Busy professionals and inventory managers didn't face any problems while scanning the item. However, disabled people encountered some issues while scanning; some barcodes weren't detected by the scanner.
- This issue might occur because the barcode isn't aligned properly for the scanner. For smaller items, we can adjust their position to align them properly, but it can be more challenging for larger and heavier items to be moved.
- To fix this issue, we can add a handheld scanner, allowing customers to freely choose the appropriate scanner for different items so that the barcodes can be scanned more easily.
- Besides, we noticed that three of our users are not members of the store. From this, we can observe that not every store's customers have members. Therefore, the step of verifying membership before checkout can be omitted.
- To replace this step, we suggest implementing a face recognition system for membership verification in the store. Each customer approaching the checkout machine can be automatically recognized by their face.
- If they are a member, they can enjoy member discounts and rewards points. Non-members can proceed directly to scanning items for checkout.

Task 3 - Make Payment

a. Busy Professionals

Video link : https://youtu.be/3f5Qbz2Hi3s?si=3pg8s3bCQRrXvo2b

0. Make Payment.

1. Consider the payment methods carefully and choose.

2. Choose a fast way which is e-wallet apps on the phone.

    2.1 Open e-wallet apps.

    2.2 Click the pay icon to display the payment QR code.

    2.3 Put the phone near the scanner.

    2.4 Adjust the phone to make sure the QR code is within the machines to scan.

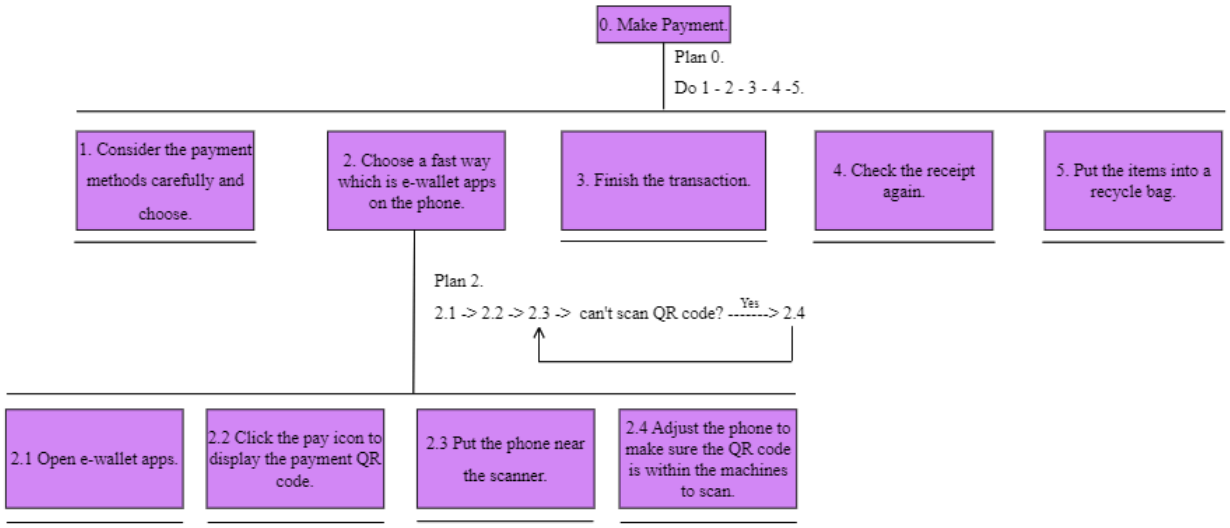
3. Finish the transaction.

4. Check the receipt again.

5. Put the items into a recycle bag.

Plan 0: Do 1, 2, 3, 4, then 5.

Plan 2: Do 2.1, 2.2, 2.3, then (2.4, if failure to scan the QR code).



b. Inventory Managers

Video link : https://youtu.be/4YzshvQbbOQ?si=sbDRzigT6uH1EMhT

0. Pay the items.

1. Select the payment method.

    1.1 Choose a credit card to pay.

2. Take out a credit card from purse.

    2.1 Put the card on the top of the machine.

    2.2 Seek for staff assistance when the card cannot be read.

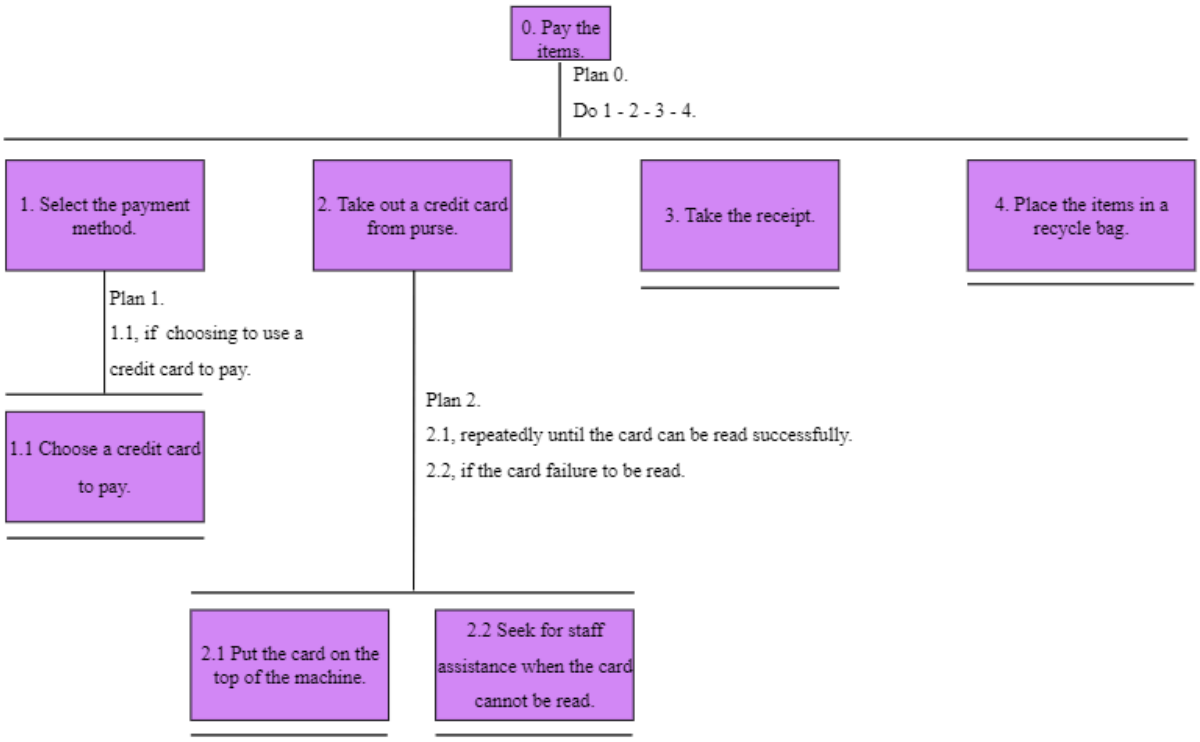
3. Take the receipt.

4. Place the items in a recycle bag.

Plan 0: Do 1, 2, 3 then 4.

Plan 1: Do 1.1 if choosing to use a credit card to pay.

Plan 2: Do 2.1(repeatedly until the card can be read successfully) and 2.2 (if the card is failure to be read)



c. Disabled People

Video link : https://youtu.be/RD2eKcE4JKc?si=PK6Gv2eSrchQusvB

0. Do Payment

1. Choose the payment method.

    1.1 Select debit card as the method.

    1.2 Insert the card into the card reader with the chip facing towards the reader.

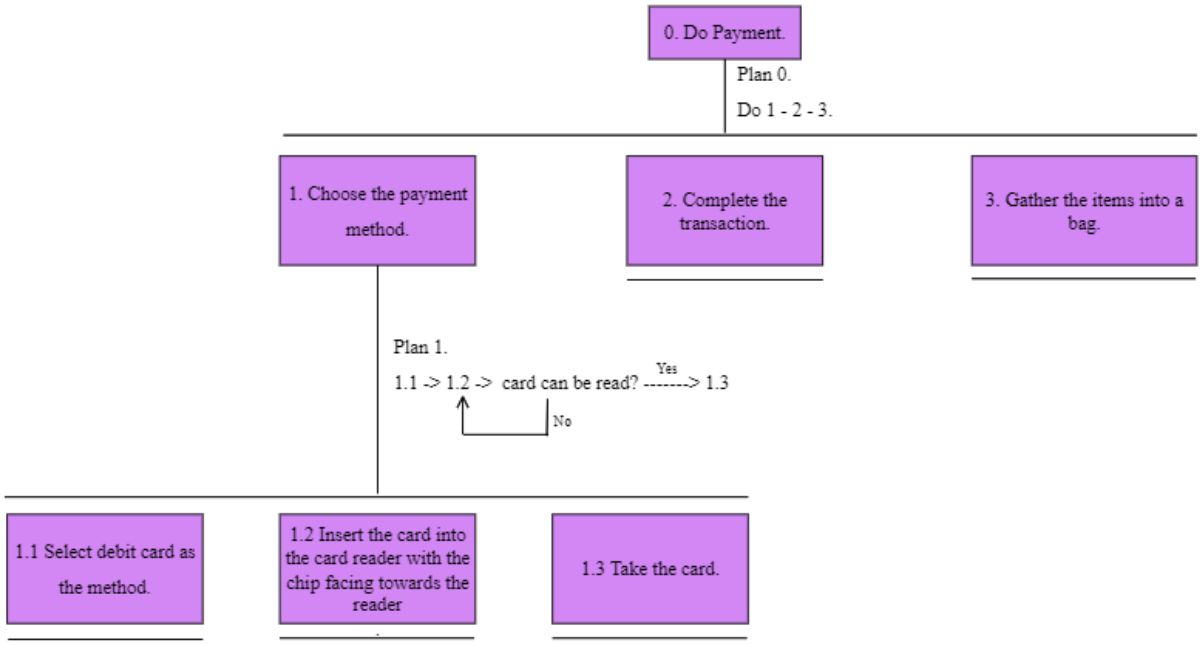
    1.3 Take the card.

2. Complete the transaction.

3. Gather the items into a bag.

Plan 0: Do 1, 2, then 3.

Plan 1: Do 1.1, (1.2 repeatedly, if the reader cannot read the card), then wait until the payment is done, do 1.3.



Findings from the HTAs from Task 3:

- During Task 1, at the payment stage, all of our users encountered some issues.
- Both the busy professional and the inventory manager experienced payment failures and had to wait for a store staff to scan their employee card before the users could select a new payment method.
- To address this issue, we suggest offering users the choice to either use self-service to cancel payments or opt for staff assistance when changing payment methods. This reduces waiting times, allowing users to quickly handle payment adjustments independently or seek help as needed, thereby saving time typically spent waiting for assistance.
- Additionally, we observed that the instructions on the screen were not clear enough for disabled people during the payment process. When attempting to make a payment, she was unsure where to place the card and hesitated before inserting it from below, which resulted in a successful payment.
- To fix this problem, we can enhance the clarity of on-screen instructions for people with disabilities. This involves adding visual cues or text prompts indicating where to place the card for payment on the main screen and directing them to insert the card into the card reader beside the machine. Besides, we can provide audio instructions or hands-on guidance for those who may have difficulty reading the screen.

Design Requirement

After observing the use of the existing system by various users, we have identified several design requirements for our new system:

- Add a handheld scanner to help users scan hard-to-reach areas, ensuring that every barcode can be scanned clearly.
- Implement face recognition functionality to verify the membership of a user.
- Offer users with the choice to either utilise self-service for cancelling payments or choose for staff assistance when changing payment methods.
- Provide a clear instruction to show user where to enter the card if they choose to pay with card

By observing the different feedback and difficulties encountered by each user during these three tasks, we have identified some issues with the existing system and formulated the design requirements mentioned above. These observations will help us create a new solution that better meets the needs of users in terms of User Interface (UI) and User Experience (UX) design.